

ELMFIRE Validation Summary Report

ELMFIRE Verification and Validation Suite

September 2, 2026

Abstract

This document aggregates the standalone reports for every runnable validation case currently discovered under `cases/Validation`. The opening summary records environment and execution configuration and reports each case's scientific decision from its case-local `outputs/metrics.json`; command completion and PDF availability are not substituted for validation evidence.

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1 Validation summary

1.1 Environment captured for this aggregate

This environment was captured when the aggregate report was generated. It describes the report-build process and must not be interpreted as the execution environment of every case unless the cases were run and aggregated in the same job. ELMFIRE is not executed to obtain this information.

Configuration	Recorded value
Captured at (UTC)	2026-09-02T17:16:45+00:00
Repository revision	6621fd3b869c
Repository state	dirty
Host platform	Linux-7.0.12-linuxkit-aarch64-with-glibc2.35
Python runtime	CPython 3.10.12 (/usr/bin/python3)
Conda environment	not active
ELMFIRE command	elmfire
ELMFIRE resolved path	/elmfire/elmfire/build/linux/bin/elmfire
ELMFIRE version declaration	not declared
ELMFIRE executable	c225b2e00349aa98128285b3d3ff9b2031d0e20944cd9f1c5ee96595d199a920
SHA-256	
MPI launcher	mpirun (Open MPI) 4.1.2
Slurm allocation	not a Slurm build job
Loaded modules	not recorded
Python packages	NumPy 1.26.4; Matplotlib 3.8.4; PyYAML 6.0.3; Rasterio 1.4.4; GDAL 3.4.1; Pandas 2.2.2; GeoPandas 1.1.4; Shapely 2.1.2

1.2 Case-declared execution configuration

These values come only from each case's `case.yaml`. They summarize the declared executable command, namelist, MPI allocation, and namelist contract. A missing declaration is reported explicitly rather than inferred.

Case	ELMFIRE command	Input namelist	MPI ranks	Schema
camp_fire	elmfire	elmfire.data.in	51	1
thomas_fire	elmfire	elmfire.data.in	51	1
tubbs_fire	elmfire	elmfire.data.in	51	1

1.3 Scientific decision summary

This table is generated from each case's case-local `outputs/metrics.json`. A successful run script or report build is not treated as scientific evidence of PASS. Missing, incomplete, blocked, or non-decisional results are reported as NOT EVALUABLE.

Total cases: 3 PASS: 0 FAIL: 0 CHARACTERIZED: 0 NOT EVALUABLE: 3

Case	Category	Title	Decision
camp_fire	landscape scale	2018 Camp Fire landscape reconstruction	NOT EVALUABLE
thomas_fire	landscape scale	2017 Thomas Fire landscape reconstruction	NOT EVALUABLE
tubbs_fire	landscape scale	2017 Tubbs Fire landscape reconstruction	NOT EVALUABLE

Decision vocabulary. PASS and FAIL are case-defined scientific decisions. CHARACTERIZED denotes descriptive validation metrics without a justified acceptance threshold. NOT EVALUABLE denotes missing, unrun, incomplete, unsupported, or otherwise non-decisional evidence.

2 Scope and decision policy

Each standalone case PDF is included intact to preserve its data provenance, configuration rationale, figures, metrics, and limitations. PASS and FAIL are used only when a case defines a justified acceptance decision. CHARACTERIZED means descriptive metrics exist without a justified pass threshold. Missing, unrun, incomplete, or otherwise non-decisional evidence is shown as NOT EVALUABLE. The linked summary at the beginning records the aggregate-build environment, case-declared execution configuration, and scientific decisions before the complete case reports.

3 Common landscape-validation evidence

The current Camp, Thomas, and Tubbs landscape cases were reconstructed from the tracked `cases/Validation/3_FIRES` archives. Each is designed to compare a 50-member ELMFIRE ensemble with cumulative VIIRS active-fire detections. The case reports retain the event-specific input provenance, namelist rationale, evidence status, and unresolved limitations.

Their common spatial metrics are Jaccard overlap, precision, recall, F_1 , predicted-to-observed area ratio, and Brier score. Together these distinguish spatial mismatch, overgrowth, undergrowth, scale bias, and probabilistic error. Because the recovered archives define no acceptance thresholds and do not quantify VIIRS-hull uncertainty, completed runs are reported as CHARACTERIZED rather than PASS or FAIL unless justified thresholds are declared before inspecting results. Unresolved model-input provenance, VIIRS collection or granule provenance, and event-specific calibration history are stated explicitly in the standalone reports.

camp_fire – 2018 Camp Fire landscape reconstruction

NOT EVALUABLE: the standalone case report PDF is missing.

thomas_fire – 2017 Thomas Fire landscape reconstruction

NOT EVALUABLE: the standalone case report PDF is missing.

tubbs_fire – 2017 Tubbs Fire landscape reconstruction

NOT EVALUABLE: the standalone case report PDF is missing.